

Yuchao Shen

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Education

Purdue University, Computer Engineering

Jan 2025 – June 2028

- GPA: 3.84/4.00
- Coursework: Data Structure, C Programming, Introduction to Digital System Design, Python Programming

Experience

Software Engineer Intern, JD.com – Beijing, China

May 2025 – Aug 2025

- Developed and maintained scalable RESTful APIs using Java and Spring Boot for internal platform services, supporting millions of daily transactions and enhancing system modularity.
- Designed and implemented real-time and batch data pipelines (Apache Flink and Spark), contributing to performance optimization in large-scale distributed systems.
- Built data consistency validation scripts using PySpark to verify the integrity of data transferred from Hive to ClickHouse (over 1B records); automated results reporting to the event monitoring center.
- Participated in production deployment and debugging, working closely with senior engineers to improve system throughput and reduce failure rates.
- Uploaded and managed pipeline jobs within JD's scheduling platform, integrating with existing data workflows for continuous delivery and monitoring.

Projects

Data Distribution and Validation System – Backend + Data Engineering Project

- Designed and implemented large-scale data migration pipelines using PySpark and SeaTunnel, supporting data synchronization from Hive to ClickHouse for datasets exceeding 1 billion records.
- Developed data validation scripts in PySpark to ensure post-transfer consistency; results were reported to internal event center systems in real time.
- Integrated pipeline jobs into the enterprise scheduling platform for production use, supporting automated execution and monitoring.
- Designed and implemented a full-stack alarm management module with CRUD functionalities using Spring Boot and Cursor, supporting real-time alerts for task failures and manual edits.
- Collaborated across backend and frontend teams to align API contracts and user interface behavior, ensuring system reliability.

AI-Enhanced Data Validation Assistant – DataHub Automation Module

- Developed an AI-powered assistant for the internal DataHub platform to support error diagnostics in large-scale data push workflows
- Integrated with a locally deployed DeepSeek-V3 large language model, using dynamically constructed prompt templates to generate natural language explanations of likely failure causes
- Streamlined incident investigation workflows and reduced human debugging effort

Technologies

Languages: C, Python, Java, SQL

Frameworks & Tools: Spring Boot, PySpark, SeaTunnel, Apache Flink, Apache Spark, Hive, ClickHouse, MySQL, Git